

Kubernetes Storage — Missing Core Concepts Addendum

1. emptyDir Volume

emptyDir is an ephemeral volume created when a Pod starts and deleted when the Pod is removed from the node.

Key behavior:

- Shared between containers in SAME pod
- Deleted when pod removed
- Survives container restart
- Lives on node storage by default

Use cases:

- Temporary cache
- Scratch space
- Inter-container file sharing

Example:

```
apiVersion: v1
kind: Pod
spec:
  containers:
    - name: app
      image: nginx
      volumeMounts:
        - mountPath: /cache
          name: cache
  volumes:
    - name: cache
      emptyDir: {}
```

Important:

emptyDir is NOT persistent storage.

2. emptyDir with Memory (tmpfs)

emptyDir can use RAM instead of disk.

Example:

```
volumes:
- name: memory-cache
  emptyDir:
    medium: Memory
```

Behavior:

- Ultra fast
- Uses node RAM
- Data lost when pod deleted
- Can cause OOM pressure

Good for:

- Session cache
- Temporary encryption buffers
- Fast scratch processing

3. hostPath Volume

hostPath mounts actual node filesystem into pod.

Example:

```
volumes:  
- name: logs  
  hostPath:  
    path: /var/log  
    type: Directory
```

This means:

Pod directly accesses node files.

Danger:

- Security risk
- Pod tied to specific node
- Bad portability

Common uses:

- Monitoring agents
- Log collectors
- kubelet access
- Device plugins

Example:

Prometheus node exporter.

4. ConfigMap Volumes

ConfigMaps inject configuration files.

Example:

```
volumes:  
- name: config  
  configMap:  
    name: app-config
```

Mounted files are automatically generated.

Behavior:

- Read-only
- Not for large data
- Not persistent user storage

Good for:

- app.properties
- nginx.conf
- environment configs

5. Secret Volumes

Secrets are similar to ConfigMaps but for sensitive data.

Example:

```
volumes:  
- name: secret-volume  
  secret:  
    secretName: db-password
```

Mounted into tmpfs memory.

Used for:

- Passwords
- TLS certificates
- API tokens

Important:

Base64 encoding is NOT encryption.

6. CSI Drivers

CSI = Container Storage Interface.

CSI drivers allow Kubernetes to communicate with storage systems.

Examples:

- AWS EBS CSI
- EFS CSI
- Ceph CSI
- Azure Disk CSI

Without CSI:

Kubernetes cannot dynamically provision modern storage.

Flow:

PVC → CSI Driver → Cloud API → Disk created

7. Inline Ephemeral CSI Volumes

Pods can request temporary CSI-managed storage.

Example:

```
volumes:
- name: ephemeral-volume
  csi:
    driver: example.csi.k8s.io
```

Deleted automatically with pod.

Good for:

- temporary workloads
- batch jobs
- CI/CD pipelines

8. projected Volumes

projected volumes combine multiple sources into one mount.

Example:

- ConfigMap
- Secret
- ServiceAccount token

Combined into:

/projected-data

Useful for:

- Application bootstrap
- Unified configuration

9. subPath Mounting

subPath mounts only a subdirectory/file from a volume.

Example:

```
volumeMounts:
- mountPath: /etc/mysql/my.cnf
  name: config
  subPath: my.cnf
```

Without subPath:

Entire volume mounted.

Good for:

- Single config files
- Partial mounts

10. Volume Lifecycle Deep Understanding

Container restart:

- Volume survives

Pod restart on same node:

- PVC survives
- emptyDir survives only if same pod

Pod deletion:

- emptyDir deleted
- PVC persists

PVC deletion:

Depends on reclaim policy.

Delete:

- underlying disk deleted

Retain:

- disk preserved

11. Real Production Patterns

Databases:

- StatefulSet
- RWO
- Separate PVC per pod

Shared uploads:

- Deployment
- RWX
- EFS/NFS

Temporary cache:

- emptyDir Memory

Monitoring agents:

- hostPath

Application config:

- ConfigMap volume

TLS certificates:

- Secret volume